

Bhavya Kumar

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EDUCATION

University of Waterloo

Honours Bachelor of Mathematics (Co-op)

Waterloo, ON

Sept. 2025 – Present

- **Major:** Applied Mathematics with Scientific Computing and Scientific Machine Learning.
- **Award:** President's Scholarship.
- **Relevant Coursework:** Data Types and Structures, Probability, Algorithm Design in C, Linear Algebra II, Calculus III, Functional Programming in Racket.

PROJECTS

Wyag - Git Version Control System - *Python* | **GitHub**

May. 2026

- Engineered a fully functional Git client from scratch, implementing a **Merkle DAG** data structure and a content-addressed object store using **SHA-1 hashing** and zlib compression.
- Overcame binary parsing complexities by developing a **custom parser for the Git index file** to accurately manage the staging area, track file metadata, and detect worktree modifications.
- Resolved object identity discrepancies by writing custom lexicographical sorting algorithms for tree objects, ensuring perfect compatibility with the official Git CLI across **15 core commands**.

Taskly - Kanban App with AI Assistant - *React, Supabase, Vercel* | **GitHub** | **Live Demo**

Apr. 2026

- Diagnosed and solved a **60fps scroll regression** caused by expensive CSS effects, rebuilding the HTTP layer with **Node.js native HTTPS** and passive event listeners to improve touch navigation performance.
- Eliminated cross-tab data inconsistencies by designing a centralized mutation pipeline with an undo/redo history and a **500ms-debounced Supabase sync**, maintaining a single source of truth across guest and authenticated modes.
- Integrated **Groq's DeepSeek-R1 70B** to parse natural-language commands, utilizing a regex-fallback JSON extraction method to handle unpredictable LLM outputs and resolve fuzzy task/column queries.

Arduino Collision Detection System - *Arduino, Embedded C/C++, Ultrasonic Sensors*

July 2023

- Built a **real-time collision detection system** on Arduino, wiring **ultrasonic sensors** to embedded C/C++ control logic that continuously measures obstacle distance and triggers warning and avoidance responses.
- Calibrated sensor detection thresholds and debounced noisy readings through iterative real-world testing to reduce false positives and reliably flag incoming obstacles.

EXPERIENCE & LEADERSHIP

GDGI MUN

Head of IT

In-person

Apr. 2024 – Feb. 2025

- Directed IT logistics and media operations for a **2-day** student-run Model UN conference, managing AV infrastructure, livestreaming, and registration systems across multiple committee rooms.

Clevered AI Fellowship

Student Fellow

Remote

Jan. 2024 – Feb. 2024

- Selected for a mentored AI program led by **Oxford-affiliated researcher Ken Kahn**; built a voice-activated accessibility prototype in Python using NLP-style intent matching to handle varied user commands.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, JavaScript, Racket, SQL

Frameworks & Libraries: React, Node.js, Express, Supabase, dnd-kit, REST APIs

Tools: Git, GitHub, Vercel, VS Code, Arduino IDE

ACHIEVEMENTS

Bronze Medalist: TIMO International Math Olympiad

Finalist: Northeastern University London Essay Competition (Tech & Society)